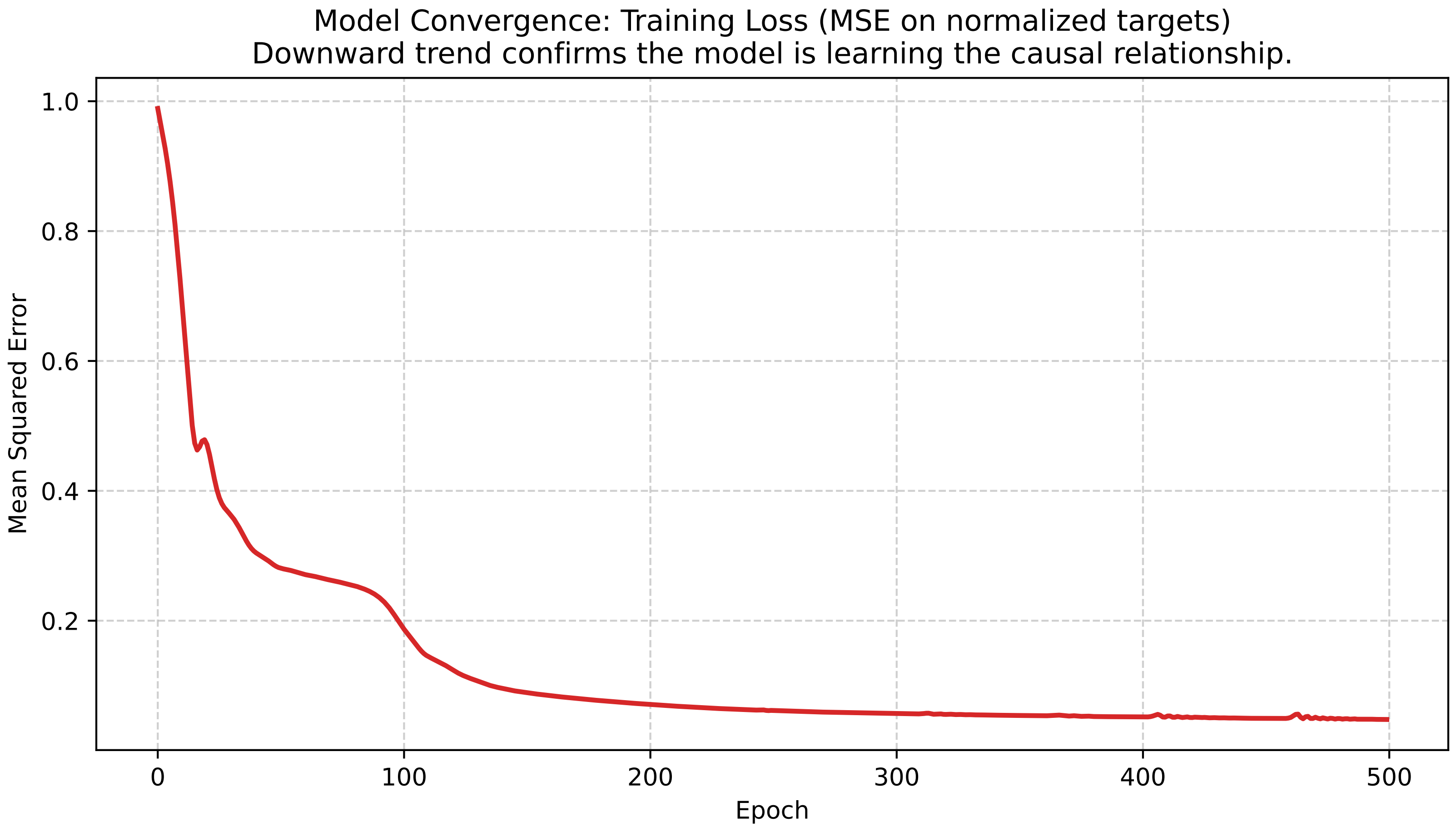
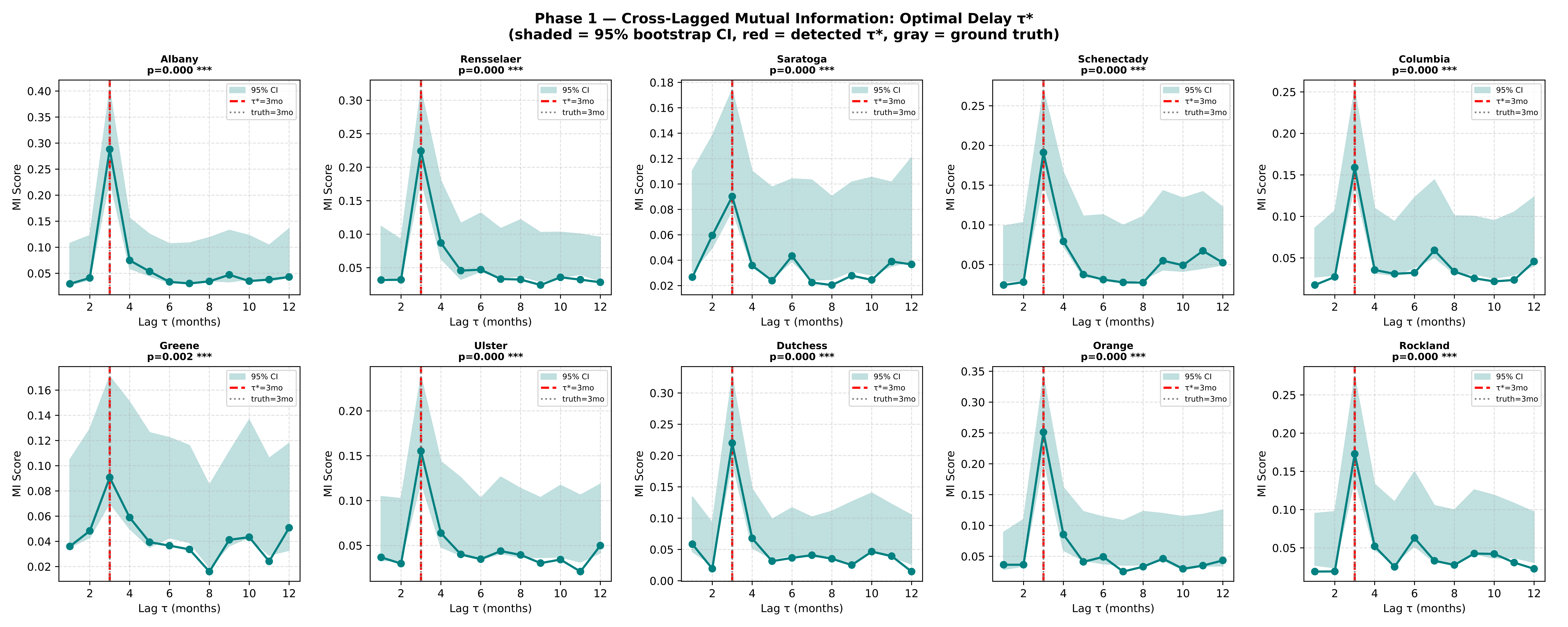
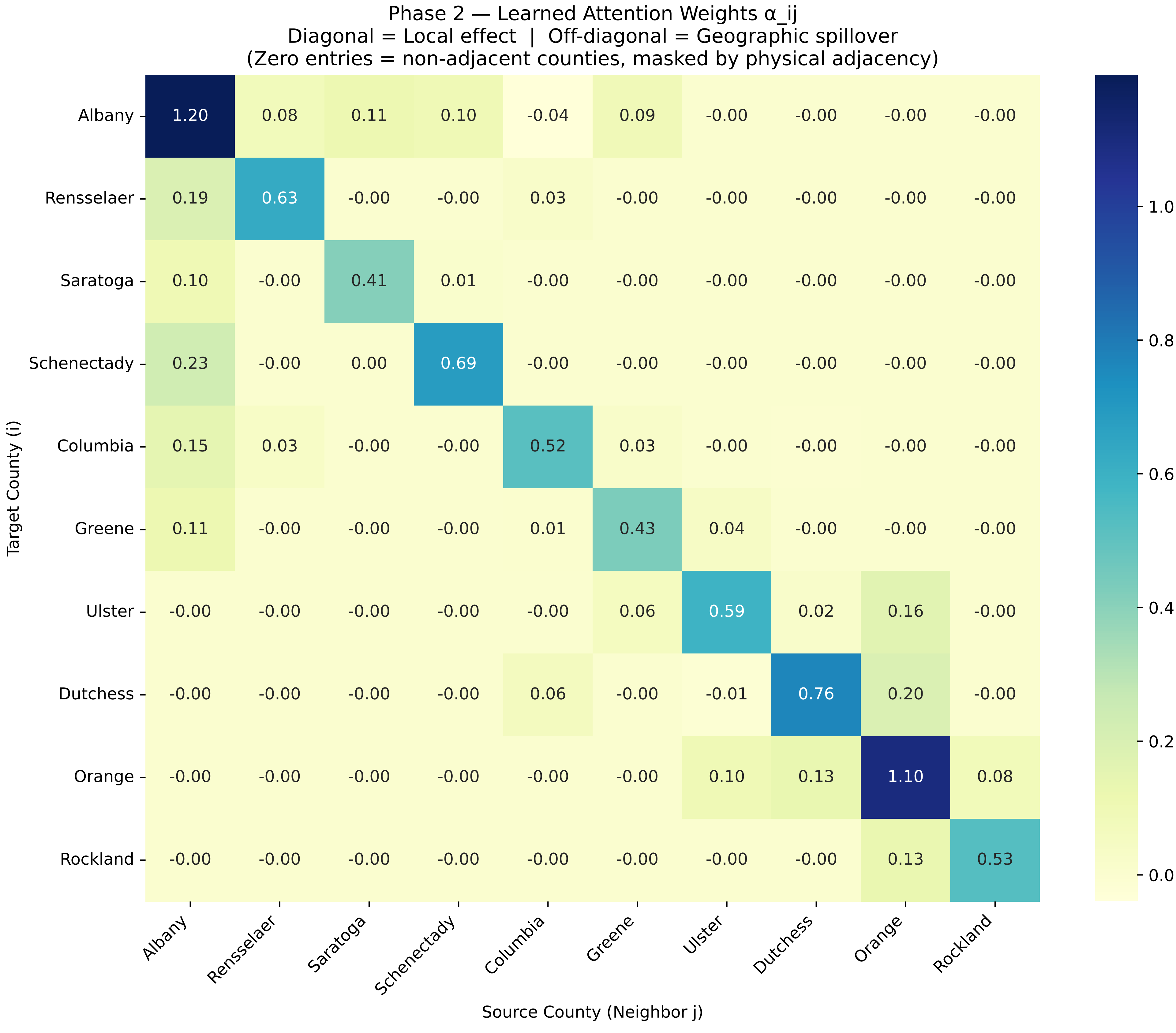
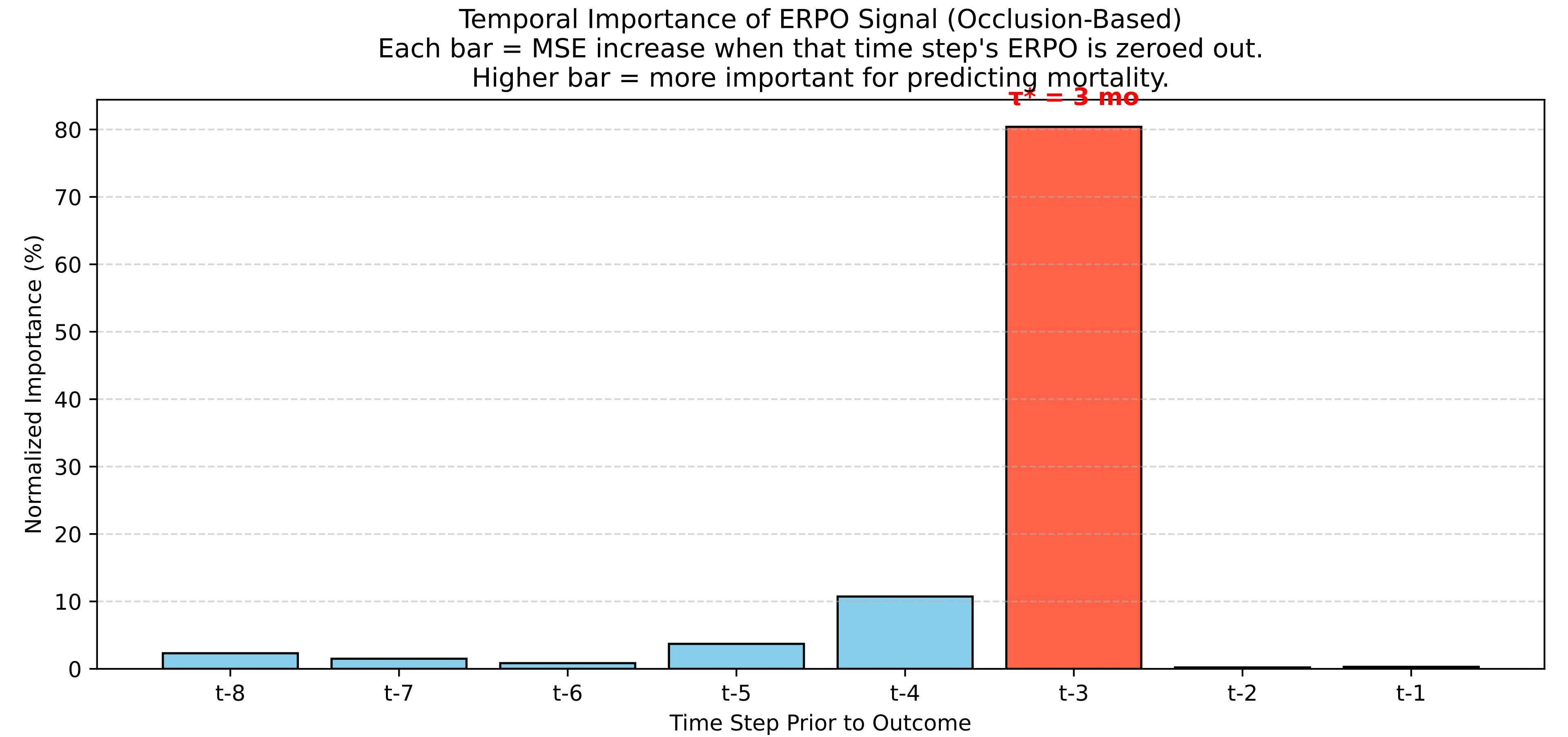
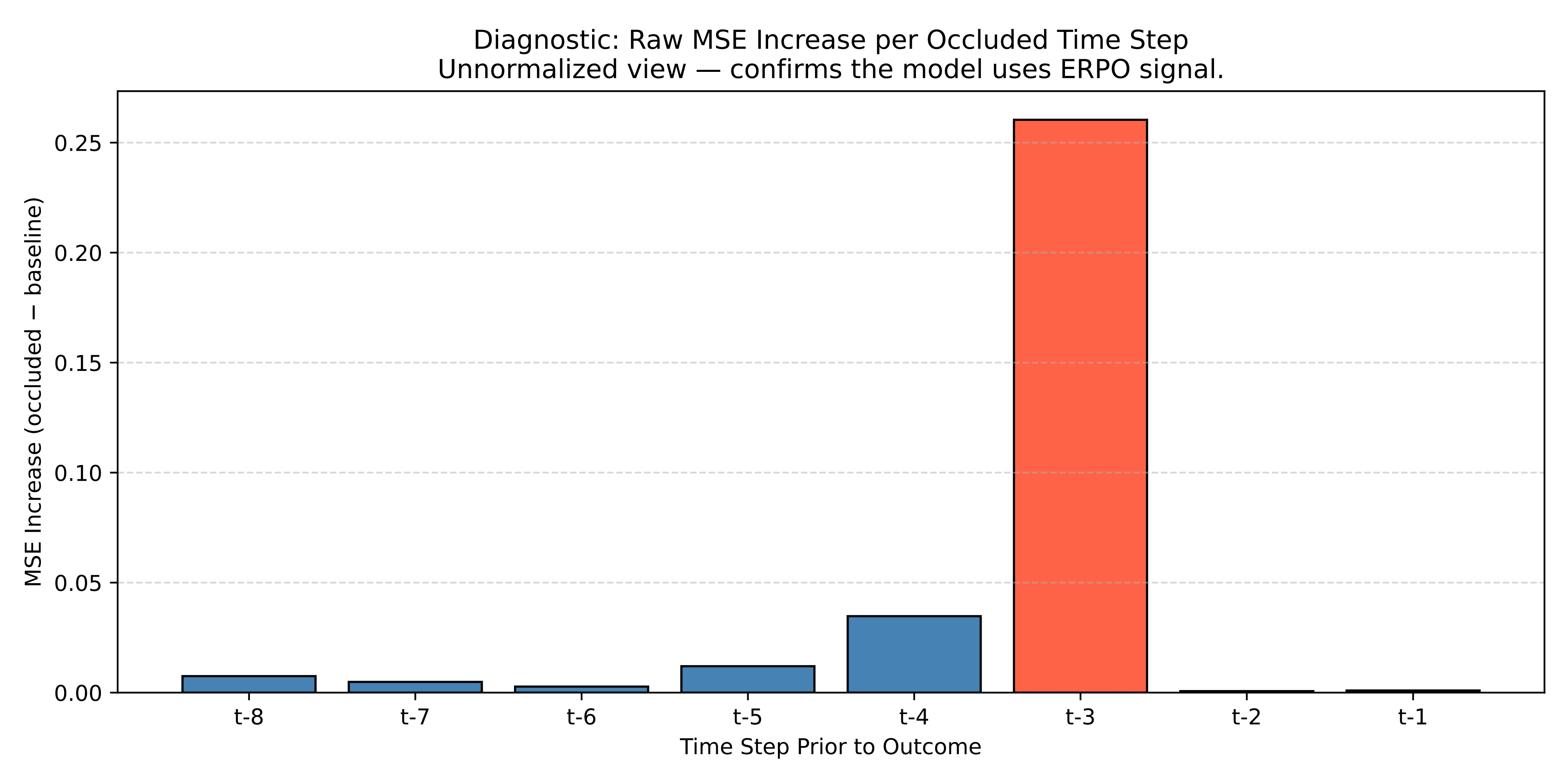

Model Convergence: Training Loss (MSE on normalized targets)
Downward trend confirms the model is learning the causal relationship.
Mean Squared Error
1.0
0.8
0.6
0.4
0.2
0
100
200
300
400
500
Epoch


Phase 1 — Cross-Lagged Mutual Information: Optimal Delay τ*
(shaded = 95% bootstrap CI, red = detected τ*, gray = ground truth)
Albany
p=0.000 ***
Rensselaer
p=0.000 ***
Saratoga
p=0.000 ***
Schenectady
p=0.000 ***
Columbia
p=0.000 ***
Greene
p=0.002 ***
Ulster
p=0.000 ***
Dutchess
p=0.000 ***
Orange
p=0.000 ***
Rockland
p=0.000 ***
95% CI
τ*=3mo
truth=3mo
MI Score
Lag τ (months)



Phase 2 — Learned Attention Weights α_ij
Diagonal = Local effect | Off-diagonal = Geographic spillover
(Zero entries = non-adjacent counties, masked by physical adjacency)
Target County (i)
Source County (Neighbor j)
Albany
Rensselaer
Saratoga
Schenectady
Columbia
Greene
Ulster
Dutchess
Orange
Rockland
1.20 0.08 0.11 0.10 -0.04 0.09 -0.00 -0.00 -0.00 -0.00
0.19 0.63 -0.00 -0.00 0.03 -0.00 -0.00 -0.00 -0.00 -0.00
0.10 -0.00 0.41 0.01 -0.00 -0.00 -0.00 -0.00 -0.00 -0.00
0.23 -0.00 0.00 0.69 -0.00 -0.00 -0.00 -0.00 -0.00 -0.00
0.15 0.03 -0.00 -0.00 0.52 0.03 -0.00 -0.00 -0.00 -0.00
0.11 -0.00 -0.00 -0.00 0.01 0.43 0.04 -0.00 -0.00 -0.00
-0.00 -0.00 -0.00 -0.00 -0.00 0.06 0.59 0.02 0.16 -0.00
-0.00 -0.00 -0.00 -0.00 0.06 -0.00 -0.01 0.76 0.20 -0.00
-0.00 -0.00 -0.00 -0.00 -0.00 -0.00 0.10 0.13 1.10 0.08
-0.00 -0.00 -0.00 -0.00 -0.00 -0.00 -0.00 -0.00 0.13 0.53
1.0
0.8
0.6
0.4
0.2
0.0


Temporal Importance of ERPO Signal (Occlusion-Based)
Each bar = MSE increase when that time step's ERPO is zeroed out.
Higher bar = more important for predicting mortality.
τ* = 3 mo
Normalized Importance (%)
0
10
20
30
40
50
60
70
80
t-8
t-7
t-6
t-5
t-4
t-3
t-2
t-1
Time Step Prior to Outcome


Diagnostic: Raw MSE Increase per Occluded Time Step
Unnormalized view — confirms the model uses ERPO signal.
MSE Increase (occluded − baseline)
0.00
0.05
0.10
0.15
0.20
0.25
t-8
t-7
t-6
t-5
t-4
t-3
t-2
t-1
Time Step Prior to Outcome

```
ERPO Spatio-Temporal Analysis — Summary Report
=======================================================

CONFIGURATION
-------------------------------------------------------
  Counties:          10
  Months:            240 (20 years)
  Sequence window:   8 months
  Training epochs:   500
  Ground truth τ:    3 months
  Spillover hubs:    Albany (idx 0), Orange (idx 8)
  Normalization:     Z-score (mean=0, std=1)

PHASE 1: CROSS-LAGGED MI — OPTIMAL DELAY τ*
-------------------------------------------------------
  County             τ*   p-value    Sig
  Albany             3mo     0.000    ***
  Rensselaer         3mo     0.000    ***
  Saratoga           3mo     0.000    ***
  Schenectady        3mo     0.000    ***
  Columbia           3mo     0.000    ***
  Greene             3mo     0.002    ***
  Ulster             3mo     0.000    ***
  Dutchess           3mo     0.000    ***
  Orange             3mo     0.000    ***
  Rockland           3mo     0.000    ***

PHASE 2: OCCLUSION-BASED TEMPORAL IMPORTANCE
-------------------------------------------------------
  Identified τ* = 3 months
  Per-step importance:
    t-8:   2.3%  (ΔMSE=0.007496)
    t-7:   1.5%  (ΔMSE=0.004865)
    t-6:   0.8%  (ΔMSE=0.002743)
    t-5:   3.7%  (ΔMSE=0.012003)
    t-4:  10.7%  (ΔMSE=0.034749)
    t-3:  80.4%  (ΔMSE=0.260384)  <-- PEAK
    t-2:   0.2%  (ΔMSE=0.000679)
    t-1:   0.3%  (ΔMSE=0.000965)

SPATIAL SPILLOVERS — Learned α_ij Weights
-------------------------------------------------------
  Albany (local α=1.20):
      ← Saratoga         α = 0.113
      ← Schenectady      α = 0.098
      ← Greene           α = 0.089
      ← Rensselaer       α = 0.077
      ← Columbia         α = -0.040
  Rensselaer (local α=0.63):
      ← Albany           α = 0.194
      ← Columbia         α = 0.026
  Saratoga (local α=0.41):
      ← Albany           α = 0.104
      ← Schenectady      α = 0.011
  Schenectady (local α=0.69):
      ← Albany           α = 0.234
      ← Saratoga         α = 0.002
  Columbia (local α=0.52):
      ← Albany           α = 0.153
      ← Rensselaer       α = 0.033
      ← Greene           α = 0.027
      ← Dutchess         α = -0.002
  Greene (local α=0.43):
      ← Albany           α = 0.114
      ← Ulster           α = 0.041
      ← Columbia         α = 0.008
  Ulster (local α=0.59):
      ← Orange           α = 0.164
      ← Greene           α = 0.058
      ← Dutchess         α = 0.020
  Dutchess (local α=0.76):
      ← Orange           α = 0.195
      ← Columbia         α = 0.063
      ← Ulster           α = -0.013
  Orange (local α=1.10):
      ← Dutchess         α = 0.134
```